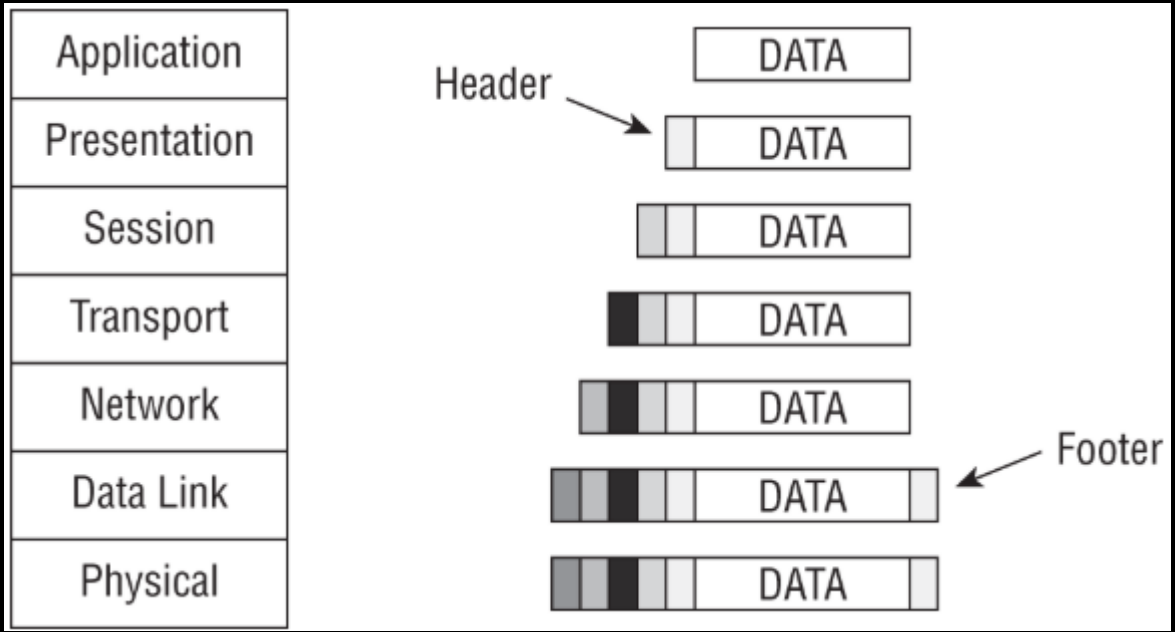
Chapter 11: Secure Network Architecture and Components

OSI Model

- Communication is only possible between protocols
- OSI History
- OSI Functionality

OSI Model

Application	7
Presentation	6
Session	5
Transport	4
Network	3
Data Link	2
Physical	1



Analyzing Network Traffic

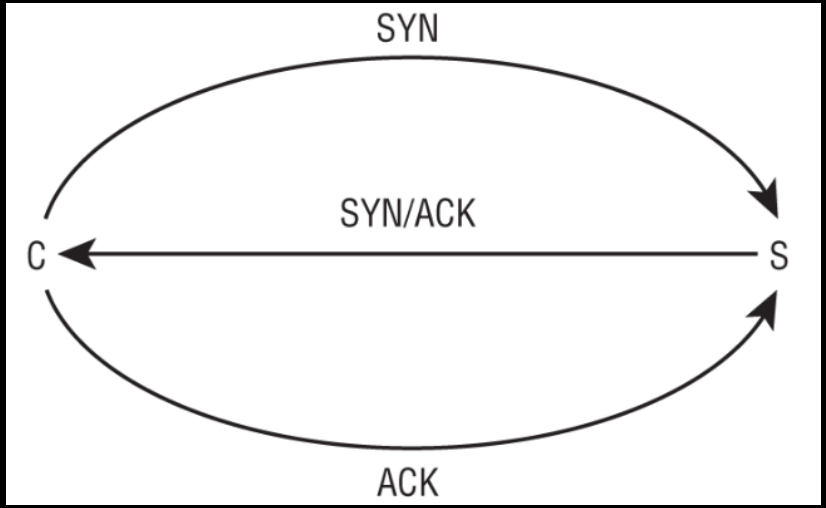
- Protocol Analyzer
- Any tools you can think of?

Common Application Layer Protocols

- Telnet, FTP, TFTP, SMTP, POP3, IMAP4, DHCP, HTTP, HTTPS, LPD, X Window, NFS, SNMP

Transport Layer Protocols

- Transmission Control Protocol (TCP)
- User Datagram Protocol (UDP)



Domain Name System

- Domain Name
- IP Address
- MAC Address
- DNS, what does it do exactly?

Domain Name System

- Top-level domain (TLD) — The com in www.google.com
- Registered domain name —The google in www.google.com
- Subdomain(s) or hostname — The www in www.google.com

Domain Name System

- DNSSEC
- DNS Poisoning
- Rougue DNS Server
- DNS Cache Poisoning
- DNS Pharming
- Altering the Hosts File

Domain Name System

- Corrupt the IP Configuration
- DNS Query Spoofing
- Use Proxy Falsification
- Defenses to DNS Poisoning?

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Internet Protocol (IP) Networking

- IPv4 vs. IPv6
- IP Classes
- Class A = /8
- Class B = /16
- Class C = /24

Internet Protocol (IP) Networking

- Internet Control Message Protocol (ICMP)
- Internet Group Management Protocol (IGMP)

ARP Concerns

- Address Resolution Protocol (ARP)
- ARP Spoofing
- Unsolicited ARP

Secure Communication

- IPSec
- Kerberos
- SSH
- Single Protocol
- Secure RPC
- TLS/SSL

Implications of Multiplayer Protocols

- TCP/IP is a multilayer protocol
- [Ethernet [IP [TCP [HTTP [Payload]]]]]
- [Ethernet [IP [TCP [TLS [HTTP [Payload]]]]]]
- [Ethernet [IPSec [IP [TCP [TLS [HTTP [Payload]]]]]]]

Convereged Protocols

- Storage Area Network (SAN)
- Internet Small Computer Systems Interface (iSCSI)
- InfiniBand over Ethernet (IBoE)
- Voice over Internet Protocol (VoIP)
- Software-Defined Networking (SDN)

Segmentation

- Bosting Network Performance
- Reducing Communication Problems
- Providing Security
- Edge Networks

Wireless Networks

- 802.11 is the IEEE standard
- Ad hoc mode
- Wireless Access Point (WAP)
- Extended Service Set Identified (ESSID)

Securing the SSID

- Service Set Identifier (SSID)
- Disable SSID broadcasting, not really securing it entirely.
- Use WPA2 or WPA3 to have encrypted channels

Wireless Channels

- 2.4GHz
- 5 GHz

Wireless Security

- Service Set Identifier (SSID)
- Disable SSID broadcasting, not really securing it entirely.
- Use WPA2 or WPA3 to have encrypted channels
- Wi-Fi Protected Setup (WPS)

Wireless Security

- Wireless MAC Filter
- Wireless Antenna Management
- Captive Portals

Wireless Communications

- Bluetooth
- Bluesniffing
- Bluejacking
- Bluesnarfing
- Bluebugging

Wireless Communications

- Radio Frequency Identification (RFID)
- Near-Field Communication (NFC)
- War-Driving
- Rouge Access Points
- Evil Twin
- Disassociation

Wireless Communications

- Jamming
- Initialization Vector (IV) Abuse
- Replay Attack
- Satellite Communications
- Cellular Networks
- Content Distribution Networks (CDNs)

Common Network Equipment

- Repeaters/Amplifiers
- Multiplexer
- Hubs
- Modems
- Bridges
- Switches

Common Network Equipment

- Routers
- LAN Extenders
- Jumpbox
- Sensor
- Collector
- Aggregators

Common Network Equipment

- Network Access Control (NAC)
- Firewalls
- Static Packet-Filtering Firewalls
- Application-Level Firewalls
- Circuit-Level Firewalls
- Stateful Inspection Firewalls

Common Network Equipment

- Next-Gen Firewalls (NGFWs)
- Internal Segmentation Firewall (ISFW)
- Proxy
- Content/URL Filter

Endpoint Security

- Transmission Media
- Coaxial Cable
- Baseband and Broadband Cables
- Twisted Pair
- Conductors
- Fiber-Optic Cables

Endpoint Security

- Transport Architecture
- Network Topologies
  - Ring Topology
  - Bus Topology
  - Star Topology
  - Mesh Topology